

COMMUNICATION PROTOCOL FOR
Vibro8 v1.0

Global Electronic Solutions D.O.O.
Cara Dušana 75
21000 Novi Sad, Serbia Tel:
+381 21 302 5993
Fax: +381 21 302 5993

E-mail: office@globalelectronic.rs

Internet: www.globalelectronic.rs

1. Document Data

Title	Communication Protocol Vibro8
Subtitle	List of messages
Document type	Communication protocol
Version	1.0

WARNINGS:

Vibro8 is a prototype for research purpose only! Read the User Manual for Vibro8!

2. Messages format

NeuraLoop will ignore wrong or incomplete messages and will respond with **>ERR<**.

Message without parameters: >XYZ<

Message consists of n ascii characters ("XYZ"), begins with ascii character ">" and ends with ascii character "<"

Message with parameters: >XYZ;xxx<

Message begins with ascii character ">" followed by n ascii characters ("XYZ") and ascii character ";" after which there are m parameters (xxx) and message ends with ascii character "<". There are four possible types of parameters:

- **\$** ascii character (one byte)
- **b** byte data type (one byte) binary value
- **w** word data type (two bytes) binary value most significant byte is at the beginning
- **l** long data type (four bytes) binary value most significant byte is at the beginning

3. Commands

Start streaming	
Command:	>0x04ST0x010x00<
Command length:	7 bytes

This command will start streaming IMU data.

Stop streaming

Command: **>0x04ST0x000x00<**

Command length: **7 bytes**

This command will stop streaming IMU data

Set vibration intensity for one vibro motor

Command: **>0x06VBlyz0x00<**

Command length: **9 bytes**

This command will set vibro motor intensity. Y represents number of the vibro motor (0x01 – 0x08) and z represents vibration intensity (0 – 100%). Vibro8 will respond with >OK<

Set vibration intensity for all vibro motors

Command: **>VBA;abcdefgh<**

Command length: **14 bytes**

This command will set vibration intensity for all vibration motors.

a – vibration intensity for vibro motor 1: 0 – 100%

b – vibration intensity for vibro motor 2: 0 – 100%

c – vibration intensity for vibro motor 3: 0 – 100%

d – vibration intensity for vibro motor 4: 0 – 100%

e – vibration intensity for vibro motor 5: 0 – 100%

f – vibration intensity for vibro motor 6: 0 – 100%

g – vibration intensity for vibro motor 7: 0 – 100%

h – vibration intensity for vibro motor 8: 0 – 100%

Set vibration duration for one motor

Command: **>0x06VBDyz0x00<**

Command length: **9 bytes**

This command will set vibration duration for one vibro motor. Y represents duration in x100ms. If y = 0, duration is infinity.

Set activity for vibro motors

Command: **>0x05VBWy0x00<**

Command length: **8 bytes**

This command will turn on/off vibro motors. Y represents vibro motors activation. LSB represents Vibro motor 1 and MSB represents vibro motor 8.

4. Data format of the message stream

Data stream that Vibro8 sends over Bluetooth is shown in the table below. One set of data consists of 21 bytes. When Vibro8 is in the stream mode, >ERR< and >OK< message will not be sent to the PC. The NDOF Fusion mode is set.

No.	Value	Explanation
1.	>	Start of the message
2.	ACC_X	Accelerometer data – X axis. Two bytes in 2's complement. MSB byte first.
3.	ACC_Y	Accelerometer data – Y axis. Two bytes in 2's complement. MSB byte first.
4.	ACC_Z	Accelerometer data – Z axis. Two bytes in 2's complement. MSB byte first.
5.	GYR_X	Gyroscope data – X axis. Two bytes in 2's complement. MSB byte first.
6.	GYR_Y	Gyroscope data – Y axis. Two bytes in 2's complement. MSB byte first.
7.	GYR_Z	Gyroscope data – Z axis. Two bytes in 2's complement. MSB byte first.
8.	EUL_Heading	Euler angle data – Heading axis. Two bytes in 2's complement. MSB byte first.
9.	EUL_Roll	Euler angle data – Roll axis. Two bytes in 2's complement. MSB byte first.

Vibro8 v1.0 Communication Protocol

10.	EUL_Pitch	Euler angle data – Pitch axis. Two bytes in 2's complement. MSB byte first.
11.	Battery SOC	Battery state of charge.
12.	<	End of the message